

RAGHURAJ SINGH CHOUHAN

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Summary

Computer Science undergraduate specializing in Artificial Intelligence Machine Learning at Manipal University Jaipur. Experienced in Full-Stack MERN Development, Natural Language Processing, Machine Learning, and Software Engineering. Currently building a production-grade Food Delivery Platform featuring secure authentication, role-based access control, real-time order tracking, and scalable backend architecture. Research experience at Newcastle University in climate policy analysis using NLP techniques. Strong foundation in Data Structures Algorithms, REST API development, database systems, and problem-solving with a passion for building impactful technology solutions.

Education

Manipal University Jaipur

B.Tech, Computer Science Engineering — Artificial Intelligence & Machine Learning

Delhi Public School, Udaipur

Senior Secondary (Class XII) — PCM: 80% | Informatics Practices: 89%

2024 – Present
pursuing 3rd Year
Completed 2023

Experience

Research Intern — Natural Language Processing

Newcastle University, Singapore | Supervisor: Dr. Anurag Sharma

Present

Remote

- Conducting NLP research on **climate change policy analysis**, processing and extracting insights from large-scale unstructured text corpora.
- Applying text preprocessing pipelines (tokenization, lemmatization, named entity recognition) and statistical methods to derive policy-level patterns from government datasets.
- Collaborating in a cross-institutional research environment, contributing to ongoing academic work with international scope.

Software Development Intern — MSME, Government of India

May 2024 | 1 Month

Technology Centre, Udaipur

- Developed and deployed a responsive government website using HTML, CSS, JavaScript, and Bootstrap, improving navigation UX and mobile accessibility.
- Built an **Arduino-based obstacle-avoiding robot** integrating ultrasonic sensors and motor control algorithms, demonstrating embedded systems proficiency.
- Designed promotional digital graphics, streamlining course outreach for the MSME Technology Centre.

Projects

SpamGuard — Multilingual SMS Spam Detection

Python, Flask, NLP, TF-IDF, scikit-learn

- Engineered a **multilingual spam classifier** supporting multiple languages using TF-IDF vectorization and ML-based classification (Naive Bayes, Logistic Regression).
- Built end-to-end NLP pipeline: text cleaning, tokenization, stopword removal, feature extraction, model training, and evaluation achieving high classification accuracy.
- Deployed via a **Flask REST API** enabling real-time SMS spam detection with a responsive web interface.

Climate Policy Text Analyzer — NLP Research Tool

Python, spaCy, NLTK, pandas

- Developed text analysis scripts to parse and classify government climate policy documents, extracting key entities, sentiments, and thematic clusters.
- Implemented automated preprocessing workflows to handle large, heterogeneous document datasets efficiently.

MSME Government Portal — Responsive Web Platform

HTML5, CSS3, JavaScript, Bootstrap

- Redesigned UI architecture for a government web portal, improving responsiveness, accessibility, and cross-browser compatibility.
- Delivered deployment-ready front-end components with structured layouts and improved navigation flow.

Technical Skills

Languages: Python, C++, C, Javascript

Full Stack Development: React.js, Node.js, Express.js, MongoDB, Mongoose, Tailwind CSS, REST APIs, JWT Authentication

ML / AI: Machine Learning, Natural Language Processing, TF-IDF, scikit-learn, Text Classification, Feature Engineering

Libraries & Frameworks: Flask, NLTK, spaCy, pandas, NumPy, Bootstrap, Git/GitHub

Core CS: Data Structures & Algorithms, Object-Oriented Programming, Embedded Systems (Arduino), REST APIs

Soft Skills: Research & Analytical Thinking, Cross-functional Collaboration, Problem Solving, Technical Communication

Achievements & Leadership

Competitive Programming — 150+ Problems

- Regularly solving DSA problems on LeetCode across arrays, trees, graphs, dynamic programming, and string algorithms.
- Focused on optimizing time and space complexity — core competency for technical interviews at top-tier companies.

IEEE Student Chapter — Active Member

- Participating member of the IEEE Student Chapter at Manipal University Jaipur, engaging in technical workshops, seminars, and peer learning initiatives.